

SIH26151 â€” Final App Flow and Data Flow Specification

****Companion to:**** `SIH26151\_FINAL\_BUILD\_GUIDE.md`

****Use:**** Implement screens, APIs, adapters, persistence and operational behavior consistently.

1. Route map

```
```text
/login
/search
/entities/:entityType/:entityId
/evidence/:evidenceId
/actors/:actorId
/actors/:actorId/graph
/actors/:actorId/timeline
/links/:linkId
/cases
/cases/:caseId
/exports/:exportId
/admin/sources
/admin/audit
```
```

2. User flow detail

Search to evidence

```
```text
User submits query
-> API validates query + role + rate limit
-> canonical entity/evidence query executes
-> search projection enriches result where available
-> API returns redacted result, evidence availability and explanation state
-> user opens entity/evidence
```
```

If there are no results, the UI must state one of: `no matching evidence collected`, `source unavailable`, `search restricted by role`, `data exists but is redacted`, or `no association found`. Never imply an unsearched source is negative evidence.

Entity to candidate actor

```
```text
Entity detail
-> load observations and related CandidateLinks
-> show original and normalized forms
-> show source/capture/observation dates
-> open a candidate actor profile only when association state permits
```
```

Candidate actor to decision

```
```text
Actor profile
-> graph/timeline request constrained by date, role, depth and result limit
-> user clicks a link edge
-> evidence drawer retrieves exact CandidateLink version + EvidenceUnits
-> user accepts/rejects/defers link with mandatory note
-> API creates new decision + audit event + timeline event
-> UI refreshes current projected state
```
```

Case to export

```

```text
User adds actor/link/evidence to case
-> case stores references, not mutable copies
-> user requests export
-> API checks authorization
-> exporter snapshots exact IDs, versions, model/config hash and timestamps
-> template renders JSON/CSV/PDF
-> artifact/hash/manifest stored
-> audit event written
-> user receives allowed download
```

```

3. UI screen requirements

| Screen | Data required | Mandatory edge-case behavior |
|-----------------|--|--|
| Login | identity, role | Do not expose role-specific routes/data before authorization |
| Search | query, filters, redacted hits | State why no result appears; apply rate/result limits |
| Entity detail | normalized + original values, observations | Show collision/confusable warning for aliases; show indicator role for key/wallet |
| Actor overview | tier, score, breakdown state, limits | Show "technical association" disclosure; distinguish no evidence from conflicting evidence |
| Graph | nodes, edges, date/depth limits | Default 2 hops; progressively expand; hide rejected edges by default; show conflicts |
| Evidence drawer | exact link + evidence versions | Show hashes, dates, source, caveats, redaction state, model version and decision history |
| Timeline | events, time confidence | Differentiate source-claimed, observed and captured time; visually mark uncertain time |
| Case | references, notes, decisions | Preserve revisions and decision history |
| Export | snapshot/manifest | Lock snapshot; do not render mutable latest state mid-export |
| Admin sources | allow/block policy, expiration, limit | Blocklist wins; source expiry stops jobs; kill switch visible |
| Audit | append-only events | Users see only audit scope authorized for their role |

4. Canonical flow records

Capture

```

```json
{
 "capture_id": "cap_...",
 "source_id": "fixture_market_a",
 "url": "fixture://market-a/profile/ghostvendor",
 "mode": "fixture_replay",
 "authorization_status": "approved",
 "captured_at": "2026-09-04T10:00:00Z",
 "source_claimed_time": "2026-09-01T08:00:00Z",
 "http_status": 200,
 "content_type": "text/html",
 "sha256": "...",
 "raw_object_reference": "fixtures/market-a/ghostvendor.html",
 "status": "succeeded"
}
```

```

Evidence unit

Use the exact `EvidenceUnit` defined in the Final Build Guide. Adapters must set an independence group; for identical/reposted content, re-use the independence group even when capture IDs differ.

Candidate link response

```

``json
{
  "link_id": "lnk_...",
  "link_version": 3,
  "left_entity_id": "alias_ghostvendor",
  "right_entity_id": "alias_nightshade99",
  "state": "needs_review",
  "score": 0.63,
  "tier": "possible_association",
  "score_status": "observed",
  "category_breakdown": {
    "K": {"score": 0.0, "state": "not_available", "evidence_ids": []},
    "I": {"score": 0.18, "state": "observed", "evidence_ids": ["ev_1"]},
    "B": {"score": 0.56, "state": "observed", "evidence_ids": ["ev_2", "ev_3"]},
    "S": {"score": 0.46, "state": "observed", "evidence_ids": ["ev_4", "ev_5"]}
  },
  "limitations": ["Text evidence is supporting only.", "Infrastructure signal may reflect shared hosting."],
  "score_model_version": "scoring-v1.0",
  "calculation_input_hash": "sha256:..."
}
...

```

5. Adapter execution flow

Common adapter rules

```

``text
CapturedDocument
-> adapter.supports(document)
-> adapter extracts candidate observations
-> adapter validates/normalizes fields
-> adapter assigns reliability, limitation, independence group and provenance
-> EvidenceUnit schema validation
-> persist canonical evidence
-> emit projection/update event
...

```

An extractor/adaptor failure must not terminate processing by other adapters. Record ``parse_failed`` or ``quarantined`` status with safe diagnostic metadata; never write raw sensitive content to logs.

Identity adapter

```

``text
Input: normalized text/profile fields
Output: PGP/wallet/contact/alias evidence
Rules: distinguish indicator observed vs verified control; wallet role required; aliases are collision-prone
...

```

Infra adapter

```

``text
Input: existing certificate matcher output
Output: certificate/twin/fingerprint evidence
Rules: add freshness, rarity and shared-hosting caveat; no high tier from infrastructure alone
...

```

OnionScan adapter

```

``text
Input: authorized scanner request + raw structured output

```

Output: OPSEC/metadata/header/analytics/certificate evidence
Rules: subprocess is non-root, isolated, time/size limited; raw output hashed; schema version stored; failure is status not crash
````

### ### MiniLM adapter

``text  
Input: cleaned alias corpora  
Output: semantic-similarity evidence  
Rules: remove templates/quotes/indicator blocks; store corpus hashes/counts/language; cap fusion contribution  
````

Classical stylometry adapter

``text
Input: eligible corpora only
Output: writing-style similarity evidence
Rules: same language; >=5 posts and >=1500 cleaned characters per alias; no output when gate fails; include feature/model version
````

### ### Behavior adapter

``text  
Input: chronologically ordered posts by alias  
Output: posting-time, vocabulary, template and source/category-overlap evidence  
Rules: standardize timezone; treat source time uncertainty explicitly; require independent non-name context for migration proposal  
````

6. Persistence and idempotency

| Object | Idempotency key | Unique rule |
|---------------|---|--|
| Capture | source + normalized URL + content hash + captured time bucket | same raw artifact may have multiple observations but must be grouped |
| Evidence | adapter source/version + capture ID + indicator type + normalized value + linked pair | duplicate insert returns existing record |
| CandidateLink | canonical ordered entity pair + score model version + calculation input hash | only one active version per exact input set |
| AuditEvent | request ID + action + object ID + timestamp sequence | append-only |
| Export | case ID + selected-version manifest hash | same request can return existing artifact |

7. Failure matrix

| Failure | API/UI response | Stored result | Retry |
|------------------------------|--|--|--------------------------|
| Source timeout | â€œSource temporarily unavailableâ€ | Capture status event | Up to 2 capped retries |
| URL not allowlisted | â€œSource blocked by policyâ€ | Blocked job/audit event | No |
| Oversized/malformed content | â€œEvidence capture quarantinedâ€ | Quarantine metadata/hash | No automatic parse retry |
| Adapter exception | â€œSome analysis unavailableâ€ | Adapter failure status | Retry only if transient |
| Scanner timeout | â€œInfrastructure scan unavailableâ€ | Scanner status/raw stderr safe reference | Optional later retry |
| Index unavailable | Search degraded; canonical profile still accessible | Projection retry job | Yes |
| Graph projection unavailable | Graph view degraded; profile/evidence still accessible | Projection retry job | Yes |
| Conflicting evidence | â€œCompeting hypothesesâ€ | Conflict set | No automatic resolution |

| Unauthorized export | â€œExport not permittedâ€ | Audit denied event | No |

8. API behavior requirements

- Every request receives a request ID and creates an audit event where relevant.
- APIs return evidence availability and redaction state, not just empty lists.
- Pagination/result limits are mandatory for search, graph and timeline.
- Dates are UTC in API responses; UI can render local timezone but must label it.
- Graph API uses date/depth/node limits and returns `truncated=true` where applicable.
- Export APIs create asynchronous jobs and return export status/artifact ID.

9. Minimal API definitions

```
```text
POST /v1/search
GET /v1/entities/{entity_type}/{entity_id}
GET /v1/evidence/{evidence_id}
GET /v1/actors/{actor_id}
GET /v1/actors/{actor_id}/graph
GET /v1/actors/{actor_id}/timeline
GET /v1/links/{link_id}
POST /v1/links/{link_id}/decision
POST /v1/cases
POST /v1/cases/{case_id}/items
POST /v1/exports
GET /v1/exports/{export_id}
GET /v1/admin/sources
POST /v1/admin/sources
POST /v1/admin/kill-switch
GET /v1/audit
```
```

10. Build checklist for every new signal

Before adding a signal, confirm:

1. What exact observation is being made?
2. What are its false-positive and false-negative modes?
3. Does it have source/capture/hash/time provenance?
4. How is it normalized and deduplicated?
5. What independence group prevents duplicate score inflation?
6. Which category K/I/B/S does it affect?
7. What maximum score/contribution applies?
8. What limitation text will the analyst see?
9. Can it create a conflict with another hypothesis?
10. Which positive, negative, stale and adversarial fixtures test it?
11. Can it work in offline fixture replay?
12. Is it permitted by source policy and safe to run?